

KARTHIK S KULAL

Artificial Intelligence & Machine Learning Engineer

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PROFILE SUMMARY

AI & Machine Learning graduate with hands-on experience building computer-vision, generative-AI, workflow-automation, and full-stack web projects using Python, OpenCV, React.js, and Next.js. Completed an AI fellowship focused on LLMs and applied generative AI. Seeking an entry-level AI/ML Engineer role to build practical, production-ready AI solutions.

EXPERIENCE

AI Fellowship Intern - LearnersByte & ExpertPedia Community

Feb 2026 – Apr 2026

Bharat Unnati NEAT 6.0 AI Fellowship

- Completed a structured fellowship covering generative AI, large language models (LLMs), and modern AI productivity tools.
- Built and presented AI-based learning exercises as part of a collaborative cohort project.
- Applied prompt-engineering techniques to solve applied AI use cases presented during the program.

PROJECTS

Pair Coder - AI-Powered Collaborative Coding & Interview Prep Platform

- Built a full-stack platform for real-time collaborative coding and technical interview practice using Next.js 15, TypeScript, Tailwind CSS, and Shadcn UI.
- Implemented live multi-user code synchronization with Socket.IO, backed by Next.js API routes / Express on the server side.
- Designed an interactive developer dashboard with Framer Motion animations for a polished, responsive user experience.
- Designed the data layer with PostgreSQL and Prisma ORM.
- Used Cloudinary for media storage and deployed the application on Vercel for production hosting.

Multi-Disease Prediction Web Application (Python, Django, Bootstrap 5, SQLite, HTML, CSS)

- Built a predictive healthcare system using machine learning to estimate the probability of various diseases.
- Applied preprocessing techniques and optimized classification models to enhance prediction accuracy.
- Developed a Django-based web application with a user-friendly interface for instant health risk analysis.
- Presented prediction outcomes through intuitive visualizations to support preventive healthcare decisions.

AI Hand Gesture Controller

- Built a real-time hand-gesture recognition system in Python (OpenCV, MediaPipe) that maps hand landmarks to system commands for gesture-based computer control.
- Tuned detection thresholds to improve recognition accuracy and reduce false positives.

Blood Donation Management System

- Developed a responsive web app (HTML, CSS, JavaScript) with donor registration and blood-group search to connect donors with people in need.

Gym Expert - Fitness Web App (React.js)

- Built a responsive React.js fitness app with a workout search feature and exercise details to help users plan training sessions.
- Focused on clean UI/UX design to improve usability across devices.

EDUCATION

Bachelor of Engineering in Artificial Intelligence & Machine Learning

Graduated 2026

Karavali Institute of Technology, Neerumarga, Mangalore (VTU) | CGPA: 8.22

SKILLS

Programming Languages: Python, JavaScript, SQL, C

Web Technologies: HTML, CSS, JavaScript, React.js, Next.js, TypeScript, Tailwind CSS, Framer Motion

Database: Node.js, Express, PostgreSQL, Prisma ORM, SQLite

AI / ML: Machine Learning, Generative AI, LLMs, Prompt Engineering, OpenCV, MediaPipe, YOLOv8

Tools & Platforms: Git, GitHub, Vercel, Cloudinary, n8n, Excel, PowerPoint, Canva, Notion AI, Word

Soft Skills: Communication, Time Management, Team Collaboration, Problem Solving

CERTIFICATIONS

- AI Fellowship - Bharat Unnati NEAT 6.0
- NPTEL / IIT Kharagpur - Privacy & Security in Online Social Media, Elite Grade (2025)
- Microsoft Azure - Introduction to the Basics of Azure Services
- AWS Application Migration Services